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A Metabolism-Driven Prognostic Model and PSMD14–SP1–GYS1 Axis Reveal Therapeutic Vulnerabilities in Melanoma

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Melanoma is a highly aggressive cutaneous malignancy characterized by a strong propensity for metastasis and therapy resistance, with its progression being closely linked to metabolic reprogramming. This study integrated multiomics data (The Cancer Genome Atlas, Gene Expression Omnibus, European Nucleotide Archive) and advanced machine learning to develop prognostic and immunotherapy prediction models for melanoma, focusing on 114 metabolism-related pathways. Cox regression identified 70 genes linked to survival, with functional enrichment revealing key metabolic pathway alterations. A metabolism-related prognostic model (MRPM) was constructed using 101 combinations of machine learning algorithms, demonstrating superior predictive accuracy across 4 cohorts. High-risk patients showed worse survival and immunotherapy response in melanoma and other cancers. Tumor microenvironment analysis revealed MRPM's negative correlation with immune infiltration and positive association with tumor purity. Single-cell sequencing highlighted MRPM gene enrichment in melanocytes. Mechanistically, *GYS1* (the key gene in MRPM) emerged as a pivotal prognostic gene that promotes melanoma proliferation and metastasis. Regulatory studies uncovered SP1's transcriptional control of *GYS1*- and PSMD14-mediated stabilization of SP1 through K48-linked ubiquitination removal. In vivo validation confirmed that PSMD14 knockdown suppressed tumor growth through SP1–*GYS1* axis disruption. This work establishes MRPM as a robust predictive tool and elucidates the PSMD14–SP1–*GYS1* regulatory network as a potential therapeutic target in melanoma metabolism.

Keywords: Deubiquitination, *GYS1*, Metabolism, Melanoma, Multiomics

INTRODUCTION

Melanoma, a highly aggressive skin cancer, has been the subject of extensive research in recent years owing to its

increasing incidence and poor prognosis, especially in advanced stage (Flaherty, 2023; Li et al, 2022; Long et al, 2023; Mueller et al, 2021). Current therapeutic options, including surgery, chemotherapy, and immunotherapy, have shown limited success, prompting the exploration of approaches (Grob et al, 2018; Ho and Sober, 1990; van Breeschoten et al, 2022). One of the major advancements in melanoma research is the identification of specific molecular markers and genetic alterations, which can aid early detection and predict patient outcomes (Amaria et al, 2022; Spencer et al, 2021; Zhou et al, 2022). Immunotherapy, particularly immune checkpoint inhibitors such as anti–PD-1 and anti–CTLA-4, has revolutionized melanoma treatment, leading to improved survival rates (George, et al, 2017; Huang and Zappasodi, 2022; Sussman and Ott, 2024). However, resistance to these therapies remains a significant challenge, underscoring the need for deeper insights into the underlying mechanisms of immune evasion and tumor progression (Dharanipragada et al, 2023). Recent studies have focused on the tumor microenvironment (TME), including immune cell infiltration, to identify therapeutic targets (Marzagalli et al, 2019). The integration of multiomics data, including genomics, transcriptomics, and proteomics, has further enhanced our understanding of melanoma's molecular landscape and paved the way for more personalized treatment strategies.

Metabolic reprogramming is a hallmark of cancer, and melanoma is no exception (Huang et al, 2023). The altered

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Abbreviations: GEO, Gene Expression Omnibus; HEK, human embryonic kidney; MRPM, metabolism-related prognostic model; PCA, principal component analysis; shRNA, short hairpin RNA; SKCM, skin cutaneous melanoma; TCGA, The Cancer Genome Atlas; TME, tumor microenvironment

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metabolism of melanoma cells plays a critical role in their growth, survival, and resistance to therapies (Tasdogan et al, 2020). Recent studies have revealed that melanoma cells often shift their metabolism from oxidative phosphorylation to aerobic glycolysis, a process known as the Warburg effect (Enríquez and Mittelbrunn, 2024; Kamenisch et al, 2016; Scott et al, 2011). This shift supports rapid cell proliferation and provides intermediates for biosynthetic pathways, such as nucleotide and lipid synthesis, crucial for maintaining cellular integrity (Scott et al, 2011). In addition, metabolic pathways, such as fatty acid oxidation, glutaminolysis, and the pentose phosphate pathway, have been implicated in melanoma progression and resistance to treatment (Pathria et al, 2018). Targeting these metabolic pathways has emerged as a promising therapeutic strategy. For instance, inhibitors of the glutamine transporter and fatty acid oxidation have shown potential in preclinical models of melanoma (Li et al, 2025; Luo et al, 2018). Understanding the metabolic changes in melanoma cells not only provides insight into the biology of the disease but also presents opportunities for the development of therapeutic agents aimed at targeting the unique metabolic features of melanoma cells.

Multomics approaches, integrating genomics, transcriptomics, proteomics, and metabolomics, are transforming melanoma research by providing a comprehensive understanding of the disease (Hu et al, 2024). RNA sequencing has been instrumental in identifying the key signaling pathways and gene expression patterns associated with melanoma progression and resistance to therapy (He et al, 2022). By combining gene expression data with clinical outcomes, researchers can develop predictive biomarkers for patient stratification and therapy selection. Proteomics has revealed post-translational modifications and protein interactions that govern melanoma cell behavior, highlighting potential therapeutic targets (Harel et al, 2019). Furthermore, metabolomics offers insights into the metabolic changes that support melanoma growth and metastasis (Davar et al, 2021). The integration of these diverse data types facilitates the identification of biomarkers, therapeutic targets, and treatment strategies. Recent advances in single-cell RNA sequencing have allowed for the examination of the heterogeneity of melanoma at the single-cell level, revealing previously unrecognized subpopulations of cells with distinct functional properties (Tirosh et al, 2016). This approach has the potential to identify therapeutic targets and improve our understanding of drug resistance mechanisms in melanoma.

This study integrated melanoma data from public databases such as The Cancer Genome Atlas (TCGA) and Gene Expression Omnibus (GEO), combined with the expression patterns of metabolism-related genes and patient prognosis information, to construct a metabolism-related prognostic model (MRPM). In addition, through the application of various machine learning algorithms, we validated the reliability of this model across different cohorts and evaluated its predictive value in immunotherapy. The study also utilized single-cell RNA-sequencing data to analyze the TME of melanoma, providing potential clues for exploring therapeutic strategies. In addition, we elucidated the PSMD14–SP1–GYS1 axis as a

therapeutic target. Our integrative approach bridges computational genomics with experimental validation, offering actionable insights for precision oncology.

RESULTS

Expression and prognostic significance of 114 metabolism-related pathways in melanoma

In a combined cohort of bulk RNA-sequencing data from melanoma (including TCGA, GSE19234, GSE22153, and GSE65904), Cox regression analysis was performed on 114 metabolism-related pathways. We identified 44 genes associated with poor prognosis (hazard ratio > 1) and 26 genes linked to improved prognosis (hazard ratio < 1), suggesting that metabolism plays a crucial role in melanoma (Supplementary Figure S1a). Data composition and batch effects were addressed as shown in Supplementary Figure S1b. The location of these genes on the chromosomes and their differential expression are shown in Supplementary Figure S1c. The red dots indicate genes that are upregulated in the disease group, whereas the black dots represent genes that are down-regulated. Kyoto Encyclopedia of Genes and Genomes and Gene Ontology analyses revealed that the prognosis-related genes were enriched in several key metabolic pathways (Supplementary Figure S1d and e). Copy number variation analysis demonstrated significant amplifications or deletions of these 114 metabolism-related pathways on chromosomes, indicating notable alterations during melanoma progression (Supplementary Figure S1f). In addition, single-sample Gene Set Enrichment Analysis scoring indicated high activation levels of all 114 metabolism-related pathways in melanoma (Supplementary Figure S1g). These findings underscore the significant alterations in metabolic genes within melanoma and their potential role in driving tumor progression.

Construction of prognostic and immunotherapy prediction models using 101 combinations of machine learning algorithms: MRPM

After identifying genes highly correlated with prognosis, we constructed the MRPM using 101 combinations of machine learning algorithms (including StepCox, random survival forest, Lasso, CoxBoost, generalized boosted regression modeling, support vector machine, etc) in TCGA skin cutaneous melanoma (SKCM) training cohorts and GEO validation cohorts. Among these, the metabolic-related prediction model built with StepCox[both] + random survival forest showed robust performance across 4 cohorts (training cohort: TCGA-SKCM; validation cohorts: GSE19234, GSE22153, and GSE65904), with an average c-index of 0.726 (Figure 1a). MRPM based on StepCox[both] + random survival forest algorithm is a prognostic and immunotherapy evaluation model composed of 14 genes, among which *GYS1* is the key gene of the model (Supplementary Figure S2a). The importance value of these genes in the model is shown in Supplementary Table S1. Subsequently, patients in the 4 cohorts were stratified into high-risk and low-risk groups on the basis of the MRPM scores, with the high-risk group consistently showing worse prognosis (Figure 1b–e). To explore the potential of MRPM beyond prognostic assessment, we performed an exploratory analysis of its predictive value in immunotherapy response across publicly available immune

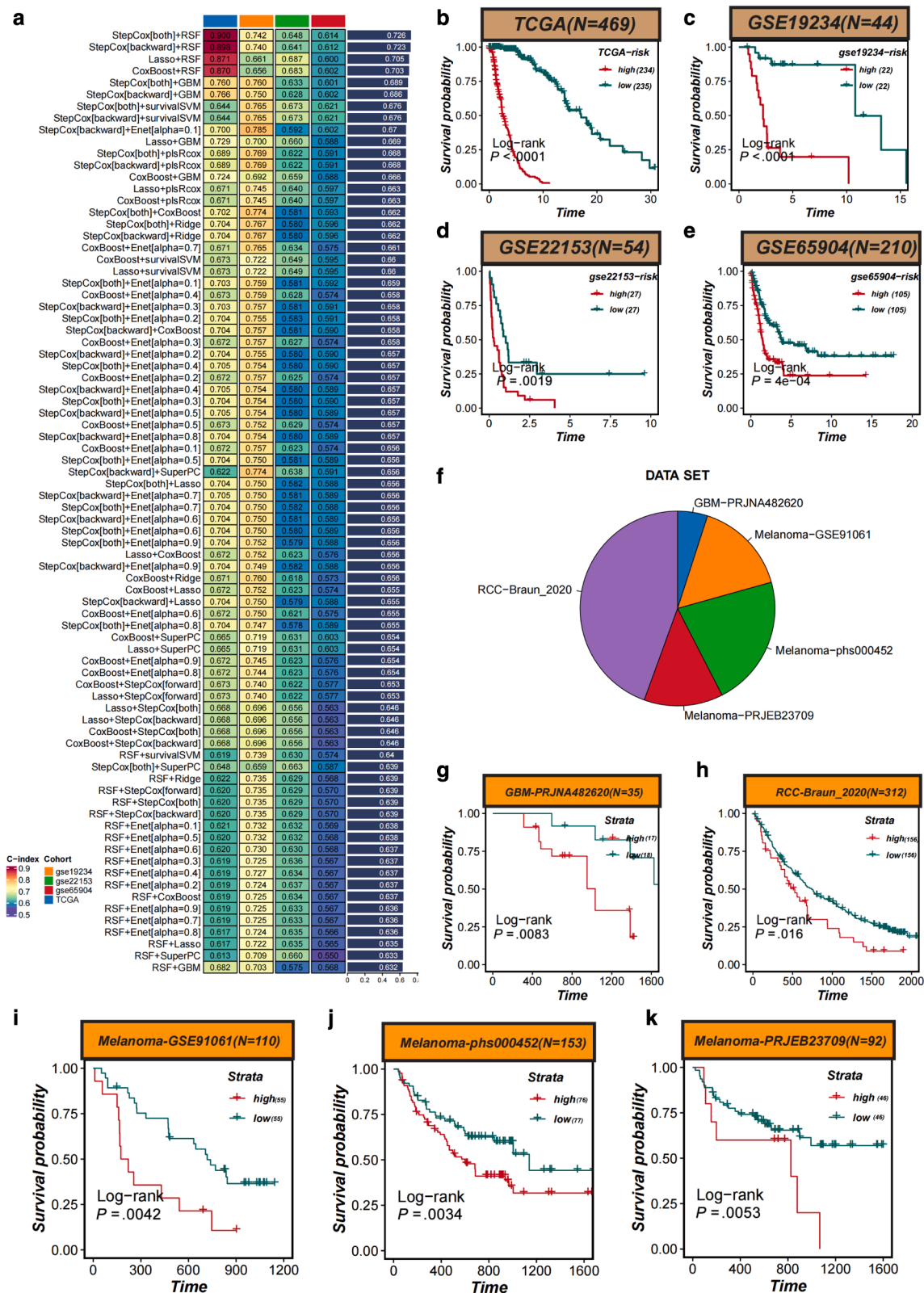


Figure 1. Construction of a MRPM using machine learning algorithms. (a) Performance ranking (C-index) of 101 combinations of machine learning algorithms; StepCox[both] + RSF achieved optimal performance across all cohorts. (b–e) Kaplan–Meier survival analyses showing significant survival differences between high-risk and low-risk groups stratified by MRPM scores in TCGA, GSE19234, GSE22153, and GSE65904 cohorts. (f) Overview of immunotherapy cohorts used for MRPM validation. (g–k) MRPM-based risk stratification predicts immunotherapy response across 5 cohorts (Braun_2020, PRJNA482620, PRJEB23709, phs000452, and GSE91061), with high-risk groups exhibiting poorer outcomes. MRPM, metabolism-related prognostic model; RSF, random survival forest; TCGA, The Cancer Genome Atlas.

checkpoint blockade datasets. As shown in Figure 1f, we included 3 melanoma immune checkpoint blockade cohorts (PEJEB23709, phs000452, and GSE91061) as well as 2 additional nonmelanoma cohorts: a renal cancer cohort (Braun et al, 2020) and a glioblastoma cohort (PRJNA482620). Notably, MRPM showed meaningful predictive performance in most melanoma immunotherapy cohorts, whereas in nonmelanoma cohorts, predictive value was observed mainly in renal cell carcinoma and glioblastoma (Figure 1g–k).

Validation of MRPM accuracy

We compared MRPM with all previously published melanoma prediction models from PubMed across 4 melanoma cohorts (TCGA, GSE19234, GSE22153, and GSE65904). Our model consistently ranked among the top, particularly in the TCGA, GSE19234, and GSE22153 cohorts (Figure 2a). To further explore the transcriptomic distinction between MRPM-defined subgroups, we performed principal component analysis (PCA) using the expression profiles of the 22 genes comprising MRPM across the 4 cohorts. As shown in Figure 2b, PCA demonstrated a clear separation between high-risk and low-risk groups, indicating that MRPM-defined patients possess distinct underlying metabolic gene expression patterns. Next, we conducted time-dependent receiver operating characteristic analysis to evaluate the predictive accuracy of MRPM at 1-, 3-, and 5-year survival endpoints across the 4 melanoma cohorts (TCGA-SKCM, GSE19234, GSE22153, and GSE65904). As shown in Figure 2c, each panel represents 1 cohort, and receiver operating characteristic curves were generated using the timeROC R package on the basis of the patient's follow-up time, survival status, and MRPM-derived risk scores. True positive fraction and false positive fraction were computed using marginal cumulative/dynamic estimators of time-dependent sensitivity and specificity. The MRPM achieved high area under the curve values across multiple time points and datasets, confirming its stable prognostic performance.

Furthermore, we conducted an analysis on the H&E-stained tissue sections of the high-risk and low-risk groups in the TCGA cohort. We also conducted further analysis on the basis of the previously published deep learning studies on H&E sections of lymphocytes (Saltz et al, 2018). The results showed that the degree of lymphocyte infiltration in the low-risk patients was significantly higher (Figure 2d).

Significance of MRPM in the TME

To explore the relationship between MRPM and the tumor immune microenvironment, we conducted a correlation analysis between MRPM scores and immune-related gene expression, including major histocompatibility complex molecules and costimulatory factors. The results demonstrated that the low-risk group exhibited significantly higher expression of immune-related molecules, suggesting enhanced immune activation (Figure 3a). Next, we performed gene set variation analysis and found that MRPM scores were negatively correlated with multiple immune-related pathways (eg, T/NK cell activation) and positively correlated with tumor-promoting pathways, such as angiogenesis and epithelial–mesenchymal transition (Figure 3b). Using the ESTIMATE algorithm, we assessed the TME and

observed that MRPM scores were negatively associated with stromal score, immune score, and ESTIMATE score, whereas they were positively correlated with tumor purity (Figure 3c), indicating an immunosuppressive and tumor-dominant microenvironment in the high-risk group. We further leveraged single-cell RNA-sequencing data to dissect the cellular heterogeneity of melanoma tissues. Clustering analysis identified several distinct cell types, including endothelial cells, NK/T cells, melanocytes, B cells, fibroblasts, and myeloid cells (Figure 3d and e). AUCCell scoring based on MRPM-related genes revealed higher activity scores in melanocytes, suggesting this cell population as a potential driver of MRPM expression (Figure 3f). Finally, we stratified all single cells into MRPM-high and MRPM-low groups and conducted cell–cell communication analysis. Interestingly, the MRPM-high group displayed markedly enhanced intercellular communication intensity, particularly among melanocytes and stromal/immune cells (Figure 3g–j), highlighting potential interactions contributing to tumor progression.

Key gene *GYS1* and its association with poor prognosis in patients with melanoma

Because *GYS1* is a key gene in the MRPM model (Supplementary Figure S2a), we further explored its significance in melanoma. First, we validated the transcriptional changes of *GYS1* in the melanoma cohort GSE98394, which revealed a significantly higher expression of *GYS1* in tumor tissues (Figure 4a). Western blotting and qRT-PCR in clinical samples also confirmed the upregulation of *GYS1* expression (Figure 4b and c) (T for tumor tissue, N for normal tissue). Given that *GYS1* encodes a glycogen synthase (a key enzyme in the glycogen synthesis process), we measured the glycogen concentration in tumor tissues and normal tissues. The results showed that the glycogen content in tumor tissues was significantly increased (Supplementary Figure S2b). Immunohistochemistry data from the Human Protein Atlas database showed that *GYS1* is highly expressed in melanoma tissues (Figure 4d). Cox regression analysis across multiple melanoma patient cohorts (GSE22153, GSE46517, TCGA_SKCM) demonstrated that high *GYS1* expression was associated with poor overall survival (Figure 4e). Kaplan–Meier survival curves also confirmed that elevated *GYS1* levels correlate with poorer survival outcomes in several melanoma cohorts, including GSE22153, GSE65904, GSE190113, and TCGA_SKCM (Figure 4f). Enrichment analysis revealed that *GYS1* is strongly positively correlated with multiple metabolic pathways, while being negatively correlated with immune pathways (Figure 4g). Immune infiltration analysis from the TIMER database showed that *GYS1* expression is positively correlated with tumor purity and negatively correlated with the infiltration of B cells, CD8+ T cells, macrophages, neutrophils, and dendritic cells (Figure 4h). Furthermore, on the basis of the cross-cohort analysis of the BEST database, the correlation landscape of *GYS1* and different types of immune cells was further provided, and a reference for the correlation between *GYS1* and drug sensitivity was also offered (Supplementary Figure S3a and b). Migration assays, including wound healing and Transwell assays, revealed a marked decrease in the migration capacity of melanoma cells after *GYS1* knockdown (Figure 4i–l and Supplementary Figure S4a and b). In addition, knockdown of *GYS1* in A375 and SK-MEL-28 melanoma cell

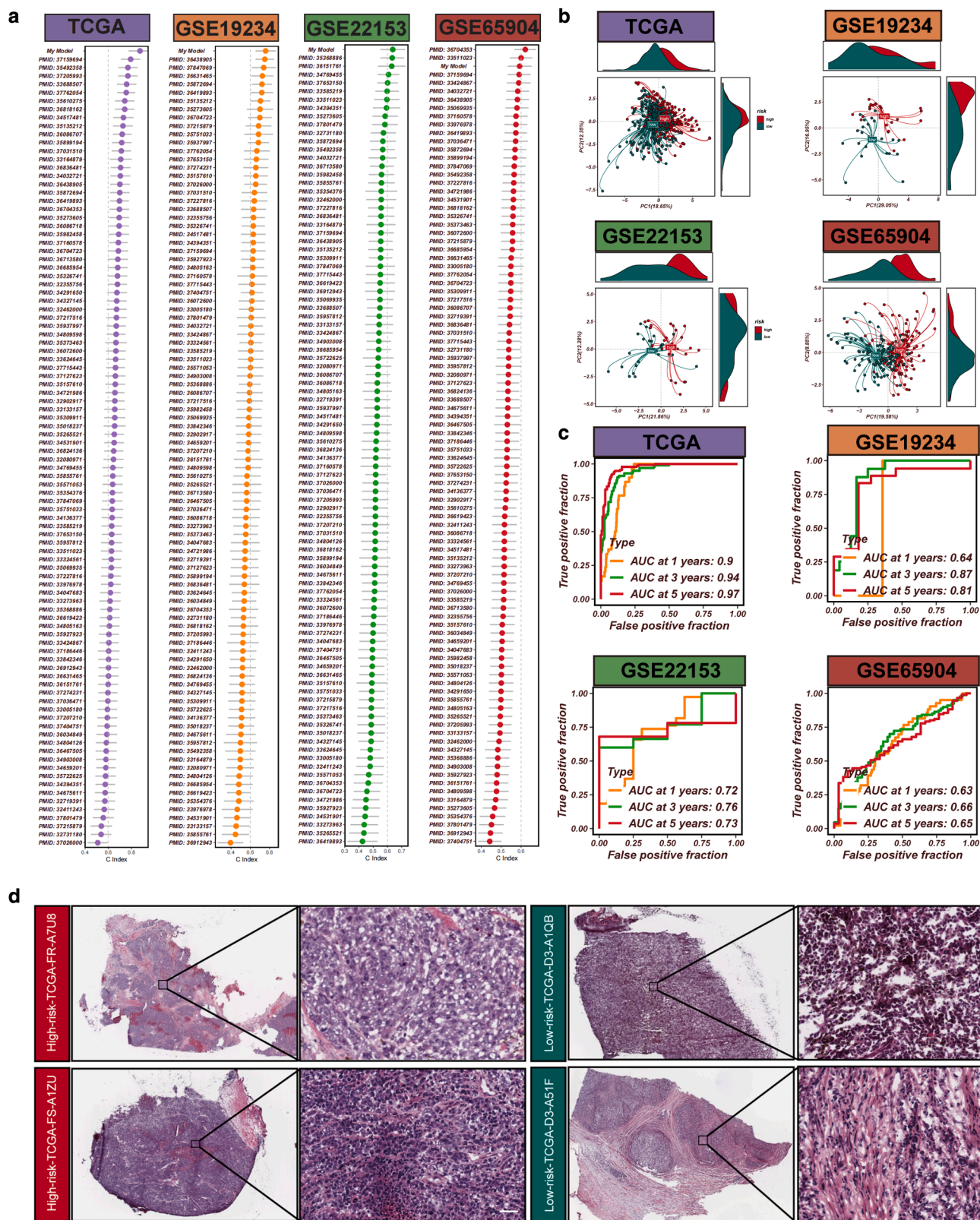


Figure 2. Validation of the accuracy and robustness of MRPM. (a) Comparative performance analysis showing MRPM ranks among the top across 4 melanoma cohorts when compared with published models. (b) PCA plots illustrating MRPM's ability to discriminate patient subgroups. (c) Time-dependent ROC curves (1, 3, and 5 years) indicating strong predictive performance of MRPM. (d) Representative H&E-stained sections from TCGA showing higher lymphocyte infiltration in the low-risk group (bar = 200 μ m). MRPM, metabolism-related prognostic model; PCA, principal component analysis; ROC, receiver operator characteristic; TCGA, The Cancer Genome Atlas.

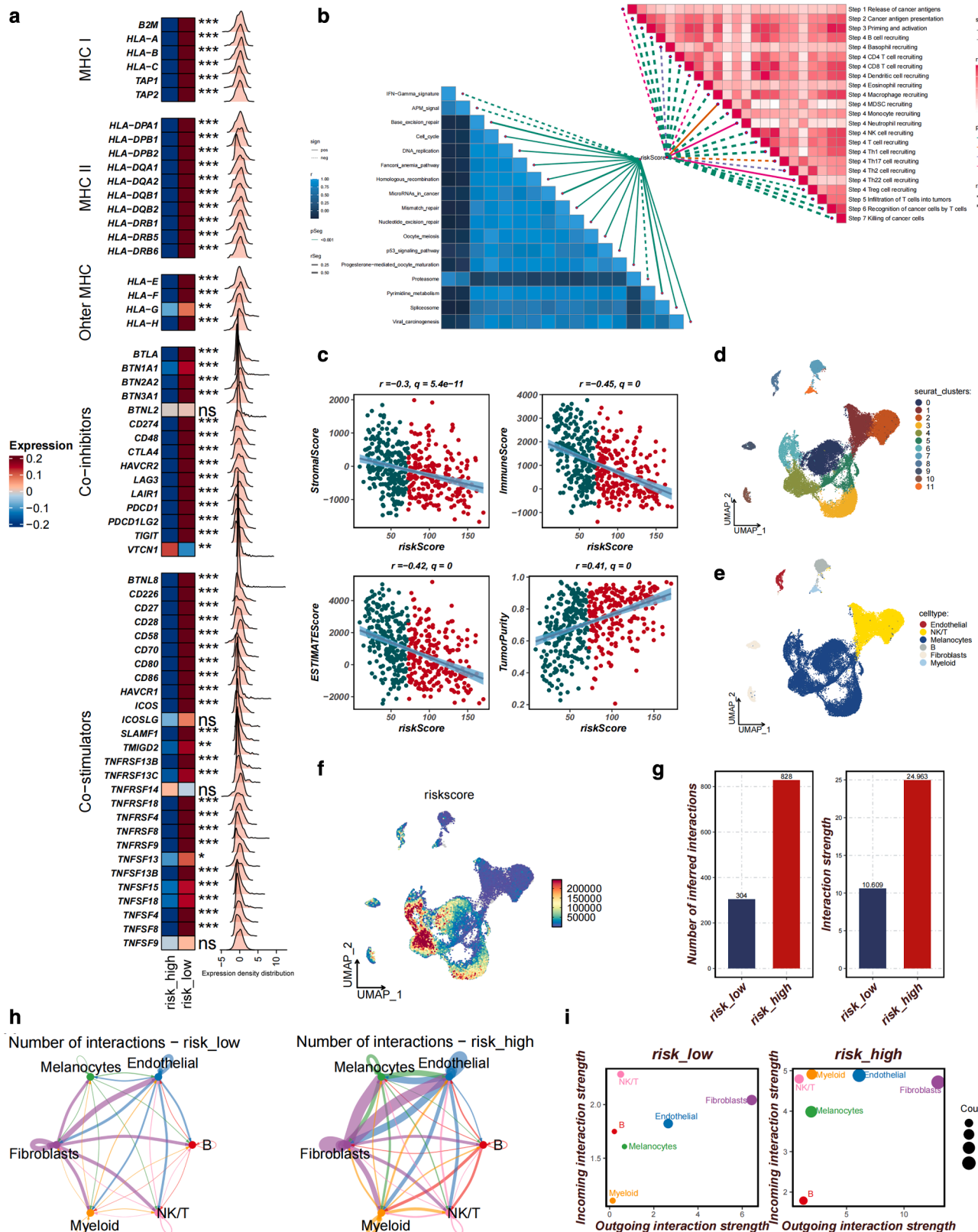


Figure 3. Role of MRPM in the tumor immune microenvironment. (a) Heatmap showing elevated expression of immune-related molecules (eg, costimulatory, MHC molecules) in low-risk group. (b) GSVA analysis revealing inverse correlation of MRPM with immune activation pathways and positive association with cancer-promoting pathways. (c) Correlation of MRPM scores with tumor microenvironment indicators: negatively correlated with stromal, immune, and ESTIMATE scores but positively with tumor purity. (d, e) UMAP and cell-type annotation from single-cell RNA sequencing of melanoma tissues. (f) AUCell scores showing MRPM-associated gene activity highest in melanocytes. (g–j) Cell–cell communication analysis indicating higher interaction intensity in the high MRPM score group. GSVA, gene set variation analysis; MHC, major histocompatibility complex; MRPM, metabolism-related prognostic model; UMAP, Uniform Manifold Approximation and Projection.

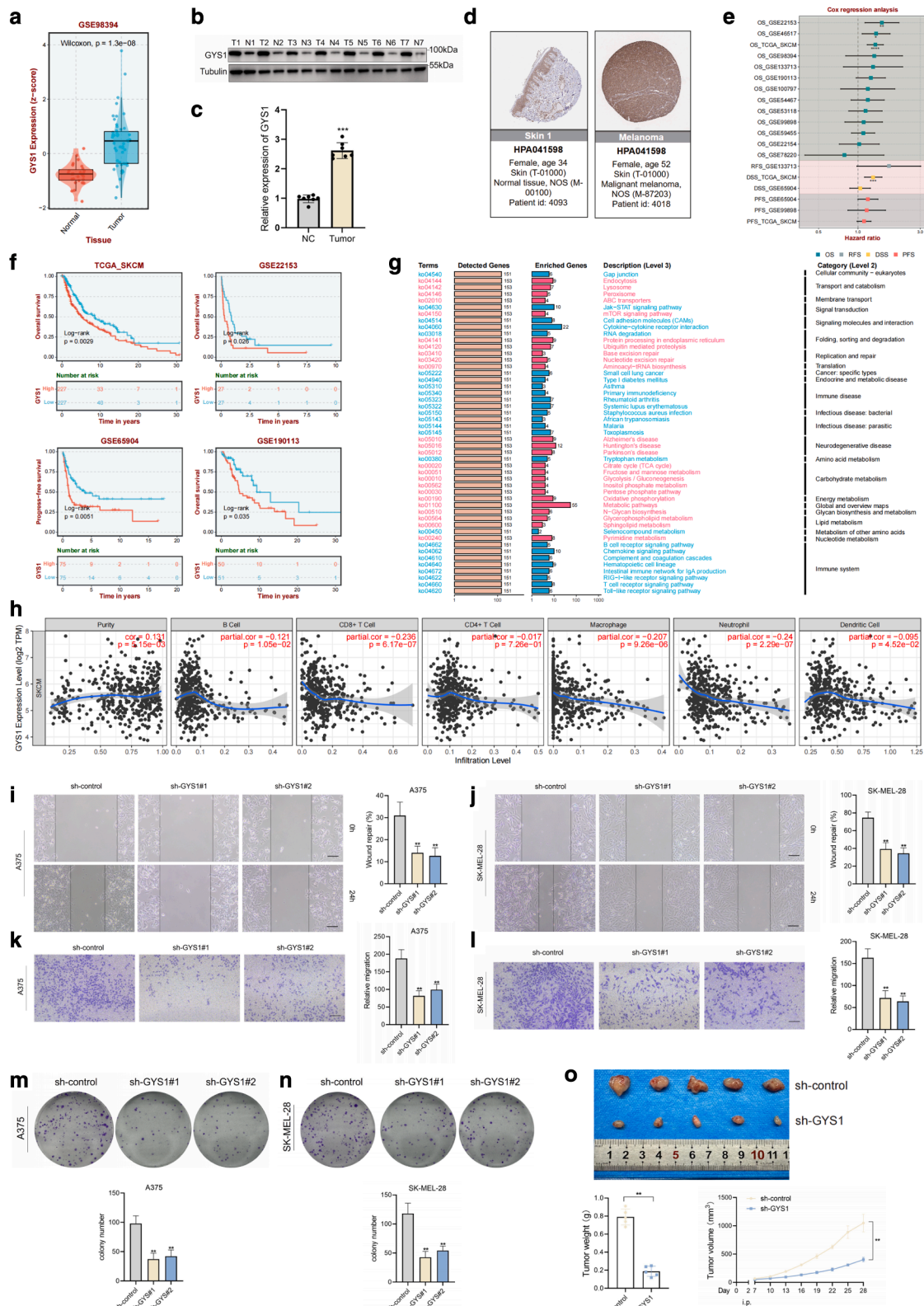


Figure 4. Functional and clinical role of GYS1 in melanoma. (a) GYS1 expression is elevated in tumor tissues from GSE98394 cohort. (b, c) Western blotting and qRT-PCR confirm GYS1 upregulation in clinical melanoma samples (T denotes tumor tissue, N denotes normal tissue). (d) IHC staining from HPA database shows high GYS1 protein levels in melanoma tissues. (e) Multivariate Cox regression analyses across several cohorts confirm GYS1 as a poor prognostic factor. (f) Kaplan–Meier curves demonstrating that high GYS1 expression correlates with poor survival in 4 independent cohorts. (g) Enrichment analysis indicates positive correlation of GYS1 with metabolic pathways and negative with immune-related pathways. (h) TIMER analysis reveals that GYS1 expression is associated with increased tumor purity and reduced immune cell infiltration. (i–l) Wound healing and Transwell assays indicate reduced migration in GYS1-silenced melanoma cells (bar = 100 μ m). (m, n) Colony-formation assays show decreased proliferation upon GYS1 knockdown. (o) In vivo xenograft model in

lines resulted in a significant reduction in cell proliferation as observed in colony formation assays (Figure 4m and n). In vivo, tumor xenografts from A375 cells in Balb/c nude mice showed significantly smaller tumor volumes and weights in the GYS1-targeted short hairpin RNA (shRNA) group than in the control-targeted shRNA group (Figure 4o).

SP1 as a positive transcriptional regulator of GYS1

To further investigate the upstream regulatory factors of GYS1, we predicted potential transcription factors for GYS1 using databases such as hTFtarget, FIMO_JASPAR, ENCODE, and GTRD and identified SP1 as one of the most significantly associated transcription factors (Figure 5a). In the JASPAR database, we predicted the SP1-binding site within the GYS1 promoter region and found that the fragment GGGGCGGGGA in the upstream –98 to –89 region is likely the binding site for SP1 (Figure 5b). Subsequently, we knocked down and overexpressed SP1 in A375 and SK-MEL-28 cell lines, observing that GYS1 expression followed a similar pattern to SP1, with expression decreasing and increasing, respectively (Figure 5c and d). qRT-PCR further confirmed these findings, showing that GYS1 expression correspondingly decreased and increased with SP1 knockdown and overexpression (Figure 5e and f). On the basis of the predicted GYS1 promoter region, we designed sequence variants (Figure 5g). Dual-luciferase reporter assays revealed that the wild-type GYS1 promoter region is essential for SP1 to enhance its transcriptional activity (Figure 5h).

PSMD14 interacts with SP1 and enhances its stability

To predict whether SP1 is regulated by deubiquitinating enzymes, we conducted a prediction of potential deubiquitinating enzymes for SP1 on Ubibrowser (http://ubibrowser.bio-it.cn/ubibrowser_v3/home/index) (Supplementary Figure S5a). Among the candidate deubiquitinating enzymes, the expression of PSMD14 was highly positively correlated with the stage, metastasis, and recurrence of melanoma (Supplementary Figure S5b–d). In A375 and SK-MEL-28 cell lines, reciprocal coimmunoprecipitation experiments demonstrated that PSMD14 and SP1 could coprecipitate together, confirming their protein–protein interaction (Figure 6a). In human embryonic kidney (HEK)293T cells, overexpression of Flag-PSMD14-WT but not its catalytically inactive mutant Flag-PSMD14-C120S dose-dependently increased the protein levels of SP1 (Figure 6b). Similarly, in HEK293T cells, cycloheximide assays revealed that Flag-PSMD14-WT but not its mutant Flag-PSMD14-C120S could enhance the protein stability of SP1 (Figure 6c). In A375 and SK-MEL-28 cell lines, knockdown of PSMD14 resulted in decreased SP1 protein levels, a phenomenon that could be rescued by treatment with MG132 (Figure 6d and 6e). In addition, in PSMD14-knockdown A375 and SK-MEL-28 cells, overexpression of Flag-PSMD14-WT but not Flag-PSMD14-C120S could restore the decreased SP1 protein levels caused by PSMD14 knockdown (Figure 6f and 6g). Cycloheximide experiments in these cell lines further showed that PSMD14 knockdown led to a significant reduction in SP1 protein stability (Figure 6h and i). These results suggest that PSMD14 interacts with SP1 and enhances its protein stability.

PSMD14 enhances SP1 protein stability by removing K48-linked ubiquitination

First, in HEK293T and PIG1 cell lines, coimmunoprecipitation experiments showed that overexpression of Flag-PSMD14-WT but not its mutant Flag-PSMD14-C120S significantly reduced SP1 ubiquitination levels (Figure 7a and b). In addition, in a cell-free in vitro system, PSMD14-WT was found to directly decrease SP1 ubiquitination, whereas PSMD14-C120S had no such effect (Figure 7c). In A375 and SK-MEL-28 cell lines, knockdown of PSMD14 led to a significant increase in SP1 ubiquitination levels (Figure 7d and e). Furthermore, PSMD14 was shown to significantly reduce K48-linked ubiquitination of SP1 (Figure 7f). In A375 and SK-MEL-28 cell lines, resistance to K48-linked ubiquitination could rescue the decrease in SP1 protein levels caused by PSMD14 knockdown (Figure 7g and h). Next, in an in vivo A375 xenograft model using Balb/c nude mice, rescue experiments were designed to validate the regulatory relationship among PSMD14, SP1, and GYS1: knockdown of PSMD14 led to reduced tumor volume and weight, but overexpression of SP1 could rescue this effect; knockdown of SP1 led to decreased tumor volume and weight, and overexpression of GYS1 could reverse this phenomenon (Figure 7i–k). The mechanistic model of this study is depicted in Figure 7l.

DISCUSSION

This study unveils the prognostic and therapeutic relevance of metabolism-related genes in melanoma, integrating multi-cohort transcriptomics, machine learning, and functional assays. The MRPM model, constructed using 101 combinations of machine learning algorithms, demonstrated remarkable accuracy in stratifying patient risk and predicting immunotherapy responses. Its generalizability across cancer types suggests that metabolic dysregulation is a conserved hallmark of tumor progression and immune evasion.

The prognostic power of MRPM stems from its incorporation of diverse metabolic pathways, including glycolysis, oxidative phosphorylation, and lipid metabolism, which are known to influence tumor aggressiveness and TME remodeling. The negative correlation between MRPM scores and immune infiltration aligns with the concept of “metabolic immunosuppression,” in which tumor cells outcompete immune cells for nutrients, impairing antitumor immunity. This is further supported by the observed downregulation of costimulatory molecules and major histocompatibility complex class II genes in high-risk groups, indicative of an immune-desert TME. The single-sample Gene Set Enrichment Analysis and TIMER2.0 analyses reinforced these findings, showing reduced CD8⁺ T cell and dendritic cell infiltration in high-risk patients, which may explain their poorer responses to immunotherapy.

The identification of GYS1 as a central driver of melanoma progression underscores the therapeutic potential of targeting glycogen metabolism (Xie et al, 2024). GYS1's role in promoting proliferation and migration, validated across in vitro and in vivo models, positions it as a biomarker and a

◀ Balb/c nude mice confirms reduced tumor growth after GYS1 knockdown. DSS, disease-specific survival; HPA, Human Protein Atlas; IHC, immunohistochemistry; OS, overall survival; PFS, progression-free survival; RFS, recurrence-free survival.

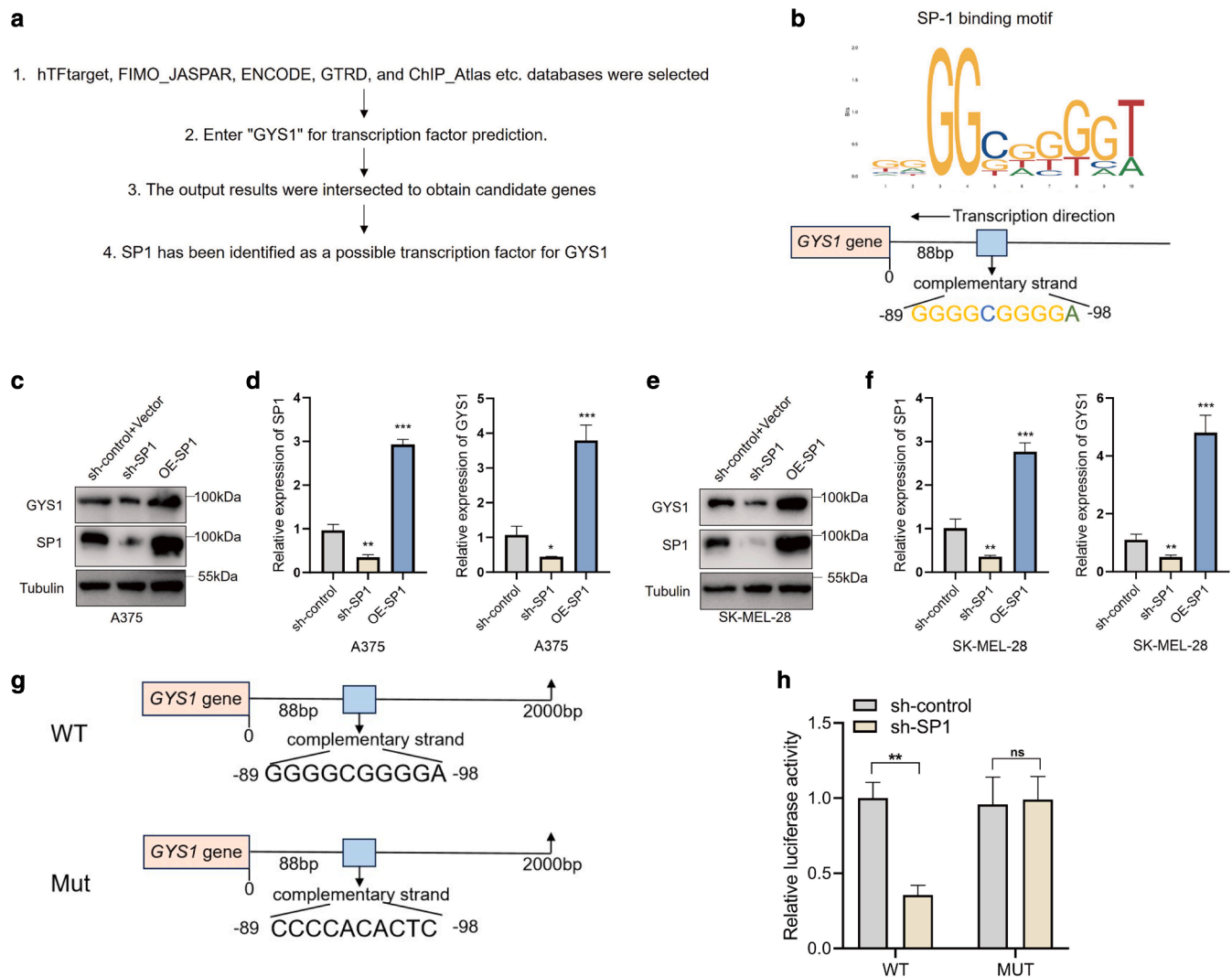


Figure 5. Transcription factor SP1 regulates GYS1 expression. (a) Venn diagram of predicted GYS1 transcription factors; SP1 is identified across multiple databases. (b) JASPAR analysis predicts SP1-binding site in GYS1 promoter region (–98 to –89 bp). (c, d) Western blotting shows GYS1 expression changes after SP1 knockdown and overexpression in A375 and SK-MEL-28 cells. (e, f) qRT-PCR validates that GYS1 mRNA levels are regulated by SP1. (g) Design of mutated GYS1 promoter construct based on predicted SP1-binding motif. (h) Dual-luciferase assays demonstrate that SP1 enhances GYS1 transcription through its promoter region.

druggable target. Its negative correlation with immune cell infiltration suggests that GYS1 inhibition could dualistically suppress tumor growth and enhance antitumor immunity, a hypothesis meriting further investigation.

Although specific GYS1 inhibitors are currently under clinical evaluation, our study sought to explore the regulatory network responsible for its aberrant overexpression in melanoma (Koch et al, 2024). This approach is grounded in the notion that upstream regulatory factors may serve as alternative or synergistic therapeutic nodes, especially in cases where resistance to direct enzyme inhibitors arises or when broader oncogenic pathways are coregulated.

Mechanistically, the discovery of the PSMD14–SP1–GYS1 axis provides insights into post-translational and transcriptional regulation of metabolic enzymes. PSMD14, a proteasome subunit, enhanced SP1 stability by deubiquitinating K48-linked ubiquitin chains, a finding consistent with prior reports of PSMD14's role in protein homeostasis. This

interaction expands the functional repertoire of SP1 beyond its canonical transcriptional role, linking proteasomal activity to metabolic reprogramming. Our rationale for investigating upstream regulators such as PSMD14 and SP1 lies in their potential to modulate not only GYS1 expression but also additional tumor-promoting genes governed by the same pathway.

Furthermore, tumor cells often exhibit 'transcriptional addiction' to specific regulators, and in such contexts, disrupting these dependencies could yield selective therapeutic advantages despite the broad cellular functions of such targets (Bradner et al, 2017). The rescue experiments in xenograft models, where SP1 or GYS1 overexpression reversed the tumor-suppressive effects of PSMD14 knockdown, validate this axis as a therapeutic vulnerability.

SP1 is a well-characterized transcription factor involved in the regulation of genes related to proliferation, apoptosis, and metabolism. Previous studies have reported that SP1

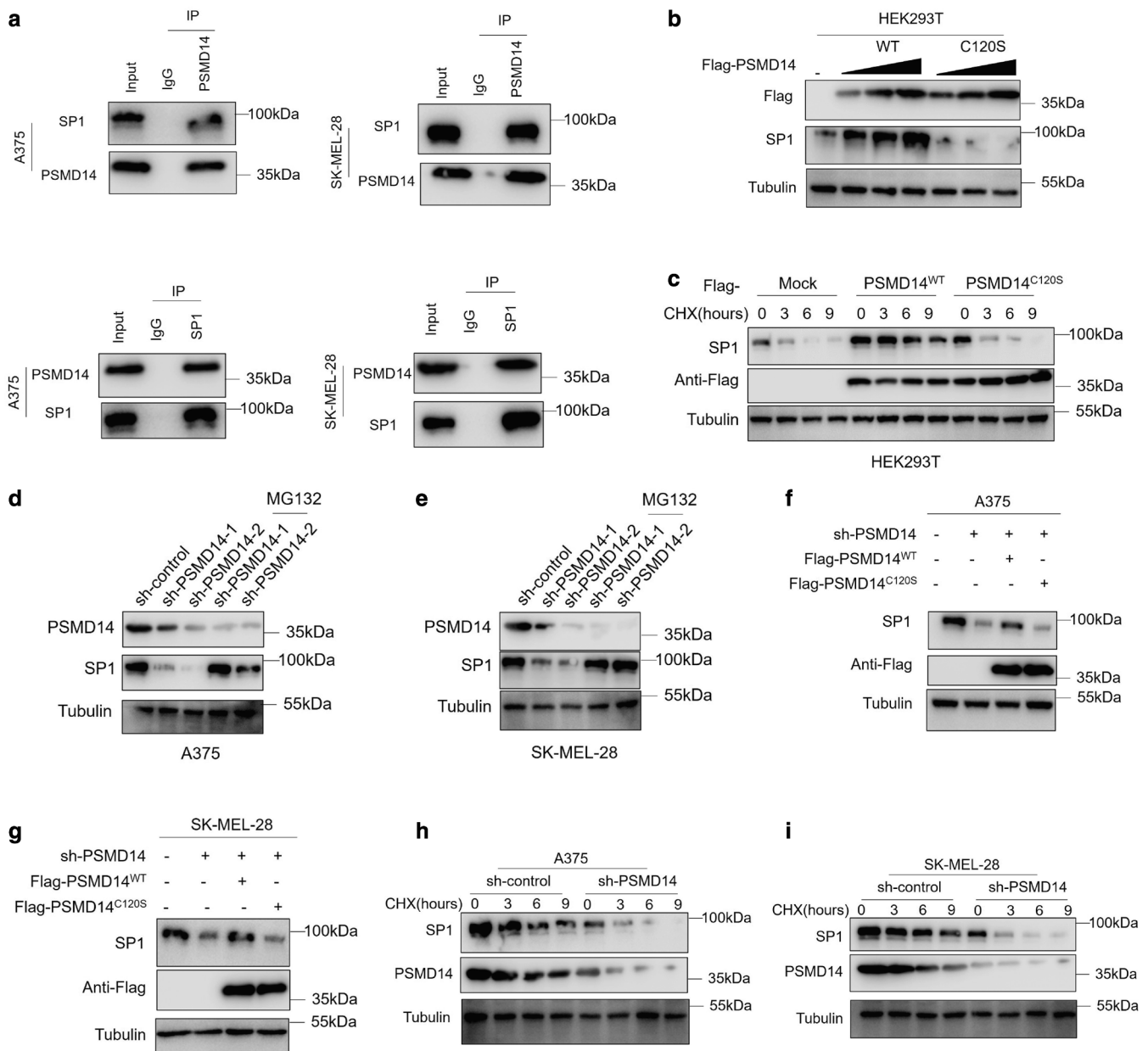


Figure 6. PSMD14 interacts with and stabilizes SP1 protein. CHX, a protein synthesis inhibitor, was used to assess SP1 protein half-life; MG132, a proteasome inhibitor, was used to determine whether SP1 degradation upon PSMD14 knockdown was proteasome dependent. (a) Co-IP experiments in melanoma cells confirm PSMD14–SP1 protein interaction. (b) Overexpression of PSMD14-WT but not catalytic mutant C120S stabilizes SP1 in HEK293T cells. (c) CHX chase assay demonstrates that PSMD14-WT prolongs SP1 protein stability. (d, e) PSMD14 knockdown reduces SP1 levels, rescued by MG132 treatment. (f, g) Rescue experiments with PSMD14-WT (but not C120S mutant) restore SP1 levels in PSMD14-depleted melanoma cells. (h, i) CHX assay shows decreased SP1 stability upon PSMD14 knockdown. CHX, cycloheximide; Co-IP, coimmunoprecipitation; HEK, human embryonic kidney.

undergoes ubiquitin–proteasome–mediated degradation, and its stability is tightly controlled through ubiquitination by E3 ligases and deubiquitination by deubiquitinating enzymes (Sun et al, 2024; Wang et al, 2025). Among the various polyubiquitin linkages, K48-linked chains are the most canonical signal for proteasomal targeting. Our results demonstrated that PSMD14 stabilizes SP1 by removing K48-linked polyubiquitin chains, thereby preventing its degradation and sustaining its transcriptional function in melanoma cells.

Despite these advances, limitations warrant consideration. First, the retrospective nature of public cohorts introduces

potential biases in clinical annotations. Prospective validation in randomized trials is needed to confirm MRPM's clinical utility. Second, although the model's pan-cancer applicability is promising, tissue-specific metabolic nuances may require customization for nonmelanoma cancers. Third, the experimental models focused on A375 and SK-MEL-28 cell lines; broader validation across melanoma subtypes (eg, BRAF mutant vs wild type) would strengthen the findings.

Future studies should explore MRPM's utility in guiding combination therapies. For instance, high-risk patients with immunosuppressive TMEs may benefit from metabolic inhibitors (eg, GYS1 antagonists) coupled with immune

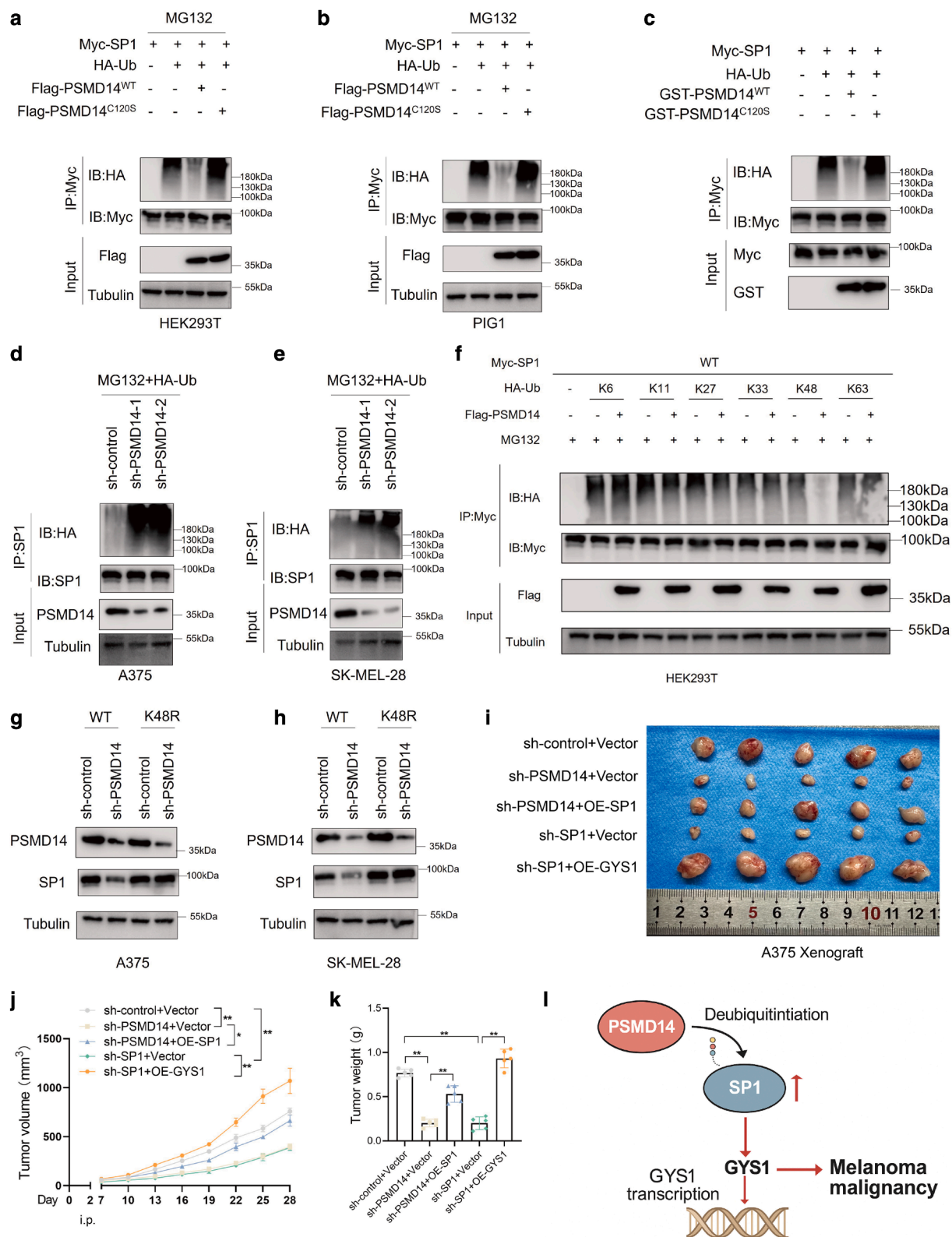


Figure 7. PSMD14 promotes SP1 stability by removing K48-linked ubiquitination. (a, b) Co-IP showing reduced SP1 ubiquitination after PSMD14-WT (but not mutant) overexpression. (c) In vitro deubiquitination assay confirms direct removal of SP1 ubiquitin chains by PSMD14-WT. (d, e) PSMD14 knockdown increases SP1 ubiquitination in melanoma cells. (f) PSMD14 preferentially reduces K48-linked polyubiquitination of SP1. (g, i) Rescue of SP1 protein levels through K48R mutant in PSMD14-depleted cells. (i–k) In vivo rescue experiments using A375 xenografts in BALB/c nude mice: PSMD14 knockdown inhibited tumor growth, which could be reversed by SP1 overexpression; SP1 knockdown reduced tumor volume and weight, and GYS1 overexpression reversed this effect. (l) Schematic model illustrating that PSMD14 stabilizes SP1 by removing K48-linked ubiquitin chains, which in turn upregulates GYS1 to promote melanoma progression. Co-IP, coimmunoprecipitation.

checkpoint blockers. In addition, the PSMD14–SP1 interaction presents an opportunity for targeted protein degradation therapies, such as PROTACs (proteolysis-targeting chimeras) (Li and Song, 2020). Single-cell and spatial transcriptomics could further dissect metabolic heterogeneity within tumor niches, informing spatially resolved therapeutic strategies.

In conclusion, this work establishes metabolism-related genes as pivotal players in melanoma progression and immune evasion. The MRPM model and the PSMD14–SP1–GYS1 axis offer dual diagnostic and therapeutic avenues, bridging computational innovation with mechanistic discovery. By contextualizing metabolic dysregulation within the TME, this study advances precision oncology paradigms for melanoma and beyond.

MATERIALS AND METHODS

Data acquisition and processing

The RNA-sequencing data of melanoma utilized in this study were obtained from publicly available databases, including TCGA and GEO. The bulk transcriptomic data comprised the TCGA-SKCM cohort and 3 independent cohorts from the GEO database: GSE19234, GSE22153, and GSE65904 (Bogunovic et al, 2009; Cirenajwis et al, 2015; Jönsson et al, 2010). Batch effects among these datasets were removed using the *sva* R package, followed by data normalization. RNA expression profiles were then integrated with corresponding patient survival information. The melanoma immunotherapy cohort PEJEB23709 was retrieved from the European Nucleotide Archive database, whereas the phs000452 cohort was sourced from the OmicsDI database. Additional immunotherapy-related cohorts, GSE100797 and GSE91061, were obtained from GEO. To assess the generalizability of the immunotherapy predictive model to other cancer types, we also included immunotherapy datasets from renal cell carcinoma (Braun, 2020) and glioblastoma (PRJNA482620). Single-cell RNA-sequencing data for patients with melanoma (GSE215120) were obtained from the GEO database. These datasets were utilized for prognostic analysis of metabolism-related genes, model construction, and validation.

Exploring the expression patterns and prognostic relevance of 114 metabolism-related pathways in melanoma

On the basis of a set of previously published sequences containing 112 types of metabolism-related genes, along with further updates to them, the final list included 114 metabolism-related pathways, totaling 3440 genes (Supplementary Table S2) (Guo et al, 2023). We applied Cox proportional hazards regression analysis in the TCGA cohort to evaluate their associations with patient survival, identifying both protective and risk factors. The hazard ratios of these genes were visualized using a forest plot. PCA was conducted to reduce dimensionality and visualize the distribution of samples across different datasets. By integrating data from public databases such as TCGA and GEO, we summarized the sample composition across datasets. Functional enrichment analyses, including Kyoto Encyclopedia of Genes and Genomes and Gene Ontology, were performed using the ClusterProfiler R package to explore the biological functions and metabolic pathways associated with the target genes. Copy number variation analysis was conducted to assess chromosomal alterations of the metabolism-related gene set, with results visualized as bar plots to distinguish different mutation types. Finally, single-sample Gene Set Enrichment Analysis was employed to calculate activity scores for various metabolic pathways, and bar plots were used to visualize the enrichment of different metabolic

categories. These analyses provide insights into the functional roles of the target genes.

Construction and validation of MRPM using 101 combinations of machine learning algorithms

After the initial Cox regression analysis to identify metabolism-related genes significantly associated with melanoma prognosis, we used the TCGA cohort as the training set and 3 independent GEO cohorts (GSE19234, GSE22153, and GSE65904) as validation sets. Diverse machine learning algorithms—including stepwise Cox regression, Lasso regression, Ridge regression, plsRcox (partial least squares Cox regression), CoxBoost, random survival forest, generalized boosted regression modeling, Enet (elastic Net), SuperPC (supervised PCA), and survival support vector machine—were applied in various pairwise combinations, resulting in a total of 101 combinations of machine learning algorithms. These models were used to construct a MRPM for patients with melanoma. Models were ranked on the basis of their C-index (concordance index), and only those with significantly high C-index values were presented. Patients in all 4 cohorts were stratified into high- and low-risk groups separately on the basis of the median risk scores generated by the models within each dataset. Kaplan–Meier survival analysis was then performed to evaluate prognostic differences between the 2 groups. To assess the predictive utility of the MRPM in immunotherapy settings, we applied it to multiple melanoma immunotherapy cohorts (PEJEB23709, phs000452, GSE100797, and GSE91061). Further external validation was conducted using immunotherapy cohorts from 2 other cancer types—renal cell carcinoma (Braun, 2020) and glioblastoma (PRJNA482620)—to evaluate the model's generalizability. In September 2023, a comprehensive literature search was conducted on PubMed to identify all published melanoma prognostic models. Only models that disclosed the gene composition and corresponding coefficients, allowing for C-index calculation, were included. Comparative robustness assessments between our MRPM and previously published models were performed in TCGA and the 3 GEO validation cohorts (GSE19234, GSE22153, and GSE65904). In addition, PCA and receiver operating characteristic curve analysis were conducted to further evaluate the performance of the MRM in stratifying patients with melanoma by prognosis.

TME analysis

To evaluate immune cell infiltration and the activity of immune-related pathways within the TME, we applied the single-sample Gene Set Enrichment Analysis algorithm. In addition, the TIMER2.0 database was utilized, which integrates the results of multiple computational frameworks to provide a comprehensive summary of immune cell infiltration in TCGA melanoma samples. For further analysis, we employed the IOBR R package, which incorporates a variety of immune infiltration estimation algorithms. This allowed us to infer the distribution and abundance of immune cell populations within tumor tissues. By leveraging this multi-algorithm approach, we aimed to gain a more comprehensive understanding of the relationship between genomic alterations and the immune response in melanoma. Furthermore, we introduced 4 key scores related to the TME: the immune score, stromal score, estimate score, and composite (fusion) score. These scores facilitate the quantification of immune cell infiltration, stromal component presence, and the overall status of the TME, thereby offering a broader perspective for investigating tumor–immune interactions.

Single-cell RNA-sequencing analysis

Single-cell RNA-sequencing data from patients with melanoma (GSE215120) were subjected to quality control, integration using the Harmony algorithm, PCA for dimensionality reduction, and further visualization through Uniform Manifold Approximation and Projection and t-distributed Stochastic Neighbor Embedding methods. Cell clusters were annotated on the basis of their marker gene expression profiles to identify specific cell types. To assess the enrichment of model genes across different cell populations, the 'AUCell' R package was used to score gene set activity in individual cells, which were subsequently divided into high- and low-score groups. Differences in intercellular communication between these 2 groups were then analyzed using the 'CellChat' R package.

Cell culture

The human melanoma cell lines A375 and SK-MEL-28, human melanocyte cell line PIG1, HEK cell line HEK-293T, and mouse melanoma cell line B16F10 were cultured under standard conditions. A375 and SK-MEL-28 cells were cultured in RPMI-1640 and DMEM media, respectively, supplemented with 10% fetal bovine serum and 1% penicillin/streptomycin. Cells were maintained at 37 °C in a humidified atmosphere containing 5% carbon dioxide. Upon reaching 70–80% confluence, cells were digested using 0.25% trypsin-EDTA and passaged accordingly. PIG1 cells, derived from human melanocytes, were cultured in EpiLife medium supplemented with epidermal GF and antibiotics under the same incubation conditions (37 °C, 5% carbon dioxide, saturated humidity). HEK-293T and B16F10 cells were cultured in DMEM medium containing 10% fetal bovine serum and 1% penicillin/streptomycin, also maintained at 37 °C and 5% carbon dioxide. These cells were similarly digested with trypsin upon reaching 70–80% confluence for subculturing. All cell lines were cultured under sterile conditions to prevent contamination.

Transfection

The shRNAs used in this study—including PSMD14-targeted shRNA, SP1-targeted shRNA, GYS1-targeted shRNA, and their respective negative control shRNAs—were purchased from GeneChem (Shanghai, China). For overexpression experiments, constructs encoding Flag-tagged wild-type PSMD14 (Flag-PSMD14-WT), its catalytically inactive mutant (Flag-PSMD14-C120S), Myc-tagged SP1 (Myc-SP1), and various ubiquitin variants (HA-Ub-WT, HA-Ub-K6, HA-Ub-K11, HA-Ub-K27, HA-Ub-K33, HA-Ub-K48, and HA-Ub-K63) were employed. Transfections were performed using Lipofectamine 3000 (Invitrogen, catalog number L3000150), according to the manufacturer's instructions. After transfection, puromycin selection was applied to establish stable cell lines, ensuring the successful integration and efficient expression of the target genes.

Antibodies

The following primary antibodies were used in this study: anti-PSMD14 (ab109130, Abcam), anti-SP1 (21962-1-AP, Proteintech), anti-GYS1 (10566-1-AP, Proteintech), anti- β -tubulin (GB152667-100, Servicebio), anti-Flag (number 14793, Cell Signaling Technology), anti-Myc (number 2276, Cell Signaling Technology), anti-hemagglutinin (number 3724, Cell Signaling Technology), and anti-GST (number 2624, Cell Signaling Technology).

qRT-PCR and functional assays

qRT-PCR was performed to assess the mRNA expression levels and the knockdown or overexpression efficiency of PSMD14, SP1, and

GYS1. Total RNA was extracted using TRIzol reagent (Invitrogen), and cDNA was synthesized using a reverse transcription kit, following the manufacturer's instructions. qRT-PCR was conducted using SYBR Green Master Mix (Thermo Fisher Scientific) on a real-time PCR system. The relative expression levels of PSMD14, SP1, and GYS1 were normalized to those of β -actin as the internal control, and the $2^{-\Delta\Delta C_t}$ method was used for quantification. All reactions were performed in triplicate to ensure accuracy and reproducibility. The sequences of the qRT-PCR primers used are provided in [Supplementary Table S3](#).

Cell Counting Kit-8, colony formation, wound healing, and Transwell assays

The specific procedures for the Cell Counting Kit-8 assay, colony-formation assay, wound-healing assay, and Transwell migration and invasion assays were conducted as previously described in our earlier publications ([Xie et al, 2025](#)).

Western blotting

A375, SK-MEL-28, PIG1, and HEK-293T cells were cultured to 80–90% confluence and then lysed using RIPA lysis buffer (containing protease inhibitors). After lysis, the cell debris was removed by centrifugation (12,000 r.p.m., 4 °C, for 10 minutes), and the supernatant was collected. Protein concentration was determined using the BCA protein assay. Equal amounts of protein (20–50 μ g) were separated by SDS-PAGE on 10–12% gels. After electrophoresis, proteins were transferred to polyvinylidene fluoride membranes. After the transfer, membranes were blocked with 5% nonfat milk at room temperature for 1 hour. The membranes were then incubated with primary antibodies specific to the target proteins (eg, anti-PSMD14, anti-SP1, etc) at 4 °C overnight. After incubation, the membranes were washed 3 times with Tris-buffered saline containing 0.1% Tween-20, each wash lasting 5 minutes. Next, the membranes were incubated with horseradish peroxidase-conjugated secondary antibodies (antirabbit or antimouse IgG) for 1 hour, followed by further washing with Tris-buffered saline containing 0.1% Tween-20. Protein bands were visualized using ECL chemiluminescent substrate and captured by a chemiluminescence imaging system. Quantitative analysis of protein bands was performed using ImageJ software. All experiments were repeated 3 times to ensure the reliability of the results.

Coimmunoprecipitation

After extracting proteins using the method mentioned earlier, an appropriate amount of protein (500–1000 μ g) was mixed with preincubated specific antibodies (targeting the proteins of interest, such as anti-PSMD14, anti-SP1, etc) and incubated overnight at 4 °C to ensure antibody binding to the target protein. Protein A/G agarose beads (Santa Cruz Biotechnology or Thermo Fisher Scientific) were then added to capture the antibody-antigen complex, and the mixture was incubated for an additional 1–2 hours. After incubation, the protein-antibody-agarose complex was precipitated by low-speed centrifugation (1500 r.p.m., 4 °C, for 5 minutes) and washed 3 times with a Tris-buffered saline containing 0.1% Tween-20 buffer containing protease inhibitors, each wash lasting 5 minutes, to remove nonspecifically bound proteins. Finally, the precipitate was eluted with SDS loading buffer and subjected to heat treatment at 95 °C for 10 minutes. Western blotting analysis was performed to detect interacting proteins with the target protein. This method effectively identifies protein-protein interactions in A375,

SK-MEL-28, PIG1, and HEK-293T cells. All experiments were repeated 3 times to ensure the reliability of the results.

Protein stability assay and proteasome inhibition

To assess the effect of PSMD14 on SP1 protein stability, cycloheximide chase assays were conducted. HEK293T, A375, and SK-MEL-28 cells were transfected with the indicated plasmids or shRNAs. After 48 hours, cycloheximide (50 µg/ml, MedChemExpress, catalog number HY-12320) was added to block protein synthesis. Cells were harvested at the indicated time points (0, 3, 6, 9 hours), and protein levels were analyzed by western blotting. SP1 protein stability was evaluated on the basis of the rate of degradation over time. For proteasome inhibition experiments, cells were treated with MG132 (10 µM, Selleck, catalog number S2619) for 6 hours before harvest. This was used to determine whether SP1 degradation upon PSMD14 knockdown was dependent on the ubiquitin–proteasome system.

K48 linkage ubiquitination resistance assay

To investigate the role of PSMD14 knockdown in K48-linked ubiquitination and its effect on SP1 stability, wild-type and K48R-mutant (where lysine 48 is mutated to arginine, blocking K48-linked ubiquitination) ubiquitin plasmids were used for transfection. A375 and SK-MEL-28 melanoma cells were transfected with either the K48R-mutant or wild-type ubiquitin plasmid and cotransfected with either shRNA targeting PSMD14 (PSMD14-targeted shRNA) or a control shRNA (control-targeted shRNA). After transfection, cells were lysed using RIPA buffer to extract proteins, and western blotting was performed to detect the levels of PSMD14 and SP1 proteins. β -Tubulin was used as an internal control to ensure consistent protein input. The differences in SP1 stability between the wild-type and K48R groups, particularly under PSMD14-knockdown conditions, were compared to assess the role of K48-linked ubiquitination in the regulation of SP1 stability.

Subcutaneous tumor implantation model construction

In this study, subcutaneous tumor models were established using A375 (human-derived) cell lines. The cells were transfected with the following treatment groups: control-targeted shRNA + vector, PSMD14-targeted shRNA + vector, PSMD14-targeted shRNA + overexpressed SP1, SP1-targeted shRNA + vector, and SP1-targeted shRNA + overexpressed GYS1. The treated cell suspensions (1×10^6 cells/mouse) were injected subcutaneously into the right axillary region of Balb/c nude mice aged 6 weeks. Tumor growth was monitored, and tumor volume was measured regularly.

The experiment lasted for 4 weeks, during which the largest and smallest diameters of the tumors were measured weekly to calculate tumor volume. At the end of the experiment, all mice were humanely killed, and tumors were harvested for H&E staining and immunohistochemistry analysis to assess morphological changes, proliferation, and apoptosis within the tumors.

Statistical analysis

Statistical analyses were performed with GraphPad Prism 10.1.2, and results are presented as mean $\pm$ SD from at least 3 independent experiments. Differences between groups were assessed using either Student's *t*-test or 1-way ANOVA followed by Tukey's posthoc test. Pearson's correlation was used to evaluate the relationships between variables, with a $P < .05$ considered statistically significant. Significance levels are represented as * $P < .05$, ** $P < .01$, and *** $P < .001$.

ETHICS STATEMENT

This study was performed in accordance with the Declaration of Helsinki. Human primary cell lines included in this study were approved as part of this study protocol. Collection of human tissue samples for this study was approved as part of the study protocol. This human study was approved by Ethics Committee of the First Affiliated Hospital of Nanjing Medical University (2024-SR-949). All adult participants provided written informed consent to participate in this study. Collection of animal tissue samples for this study was approved as part of the study protocol. This animal study was approved by Institutional Animal Care and Use Committee of Nanjing Medical University (approval 2406070).

DATA AVAILABILITY STATEMENT

Datasets related to this article can be found at The Cancer Genome Atlas, hosted at The Cancer Genome Atlas-skin cutaneous melanoma (<https://portal.gdc.cancer.gov/projects/TCGA-SKCM>) and Gene Expression Omnibus, hosted at GSE215120 (<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE215120>), GSE19234 (<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE19234>), GSE22153 (<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE22153>), and GSE65904 (<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE65904>). The data that support the findings of this study are available from the corresponding author upon request.

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CONFLICT OF INTEREST

The author state no conflict of interest.

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AUTHOR CONTRIBUTIONS

Conceptualization: JX, SZ, CM, DW, MW, YH; Data Curation: JX, SZ, CM, QL, ZW, QT; Formal Analysis: JX, SZ, DW, CM, WY, PZ, QL, ZW, QT, LL, MW, YH; Investigation: JX, QT, CM, WY, PZ; Project Administration: JX, YH; Supervision: MW, YH; Writing - Original Draft Preparation: JX, DW; Writing - Review and Editing: JX, DW, YH

SUPPLEMENTARY MATERIAL

Supplementary material is linked to the online version of the paper at www.jidonline.org, and at 10.1016/j.jid.2025.08.044.

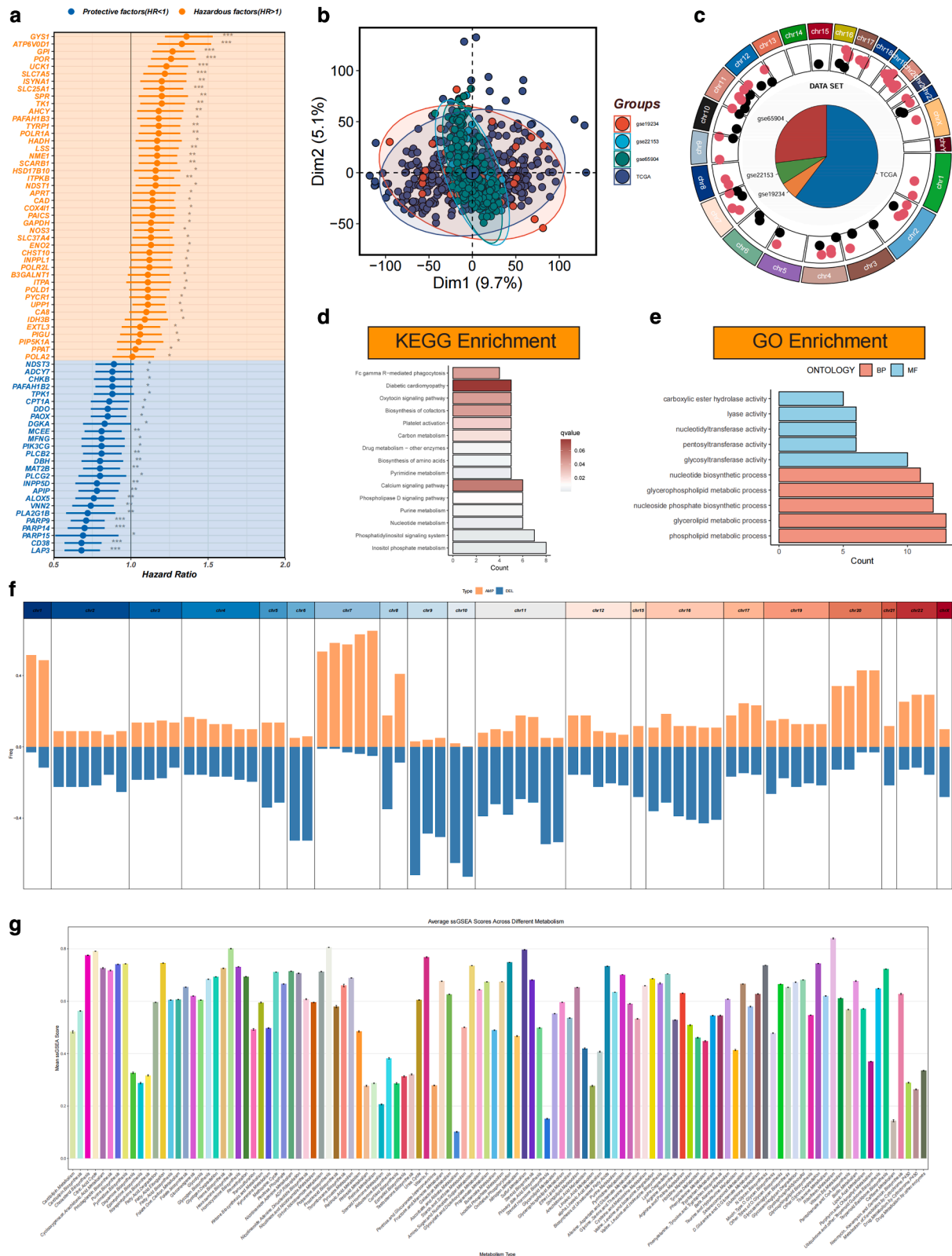
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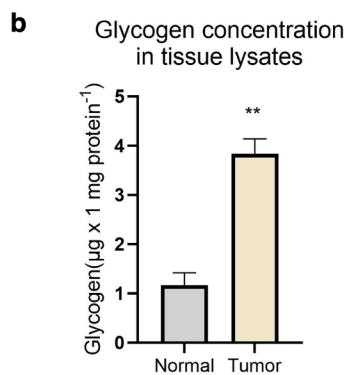
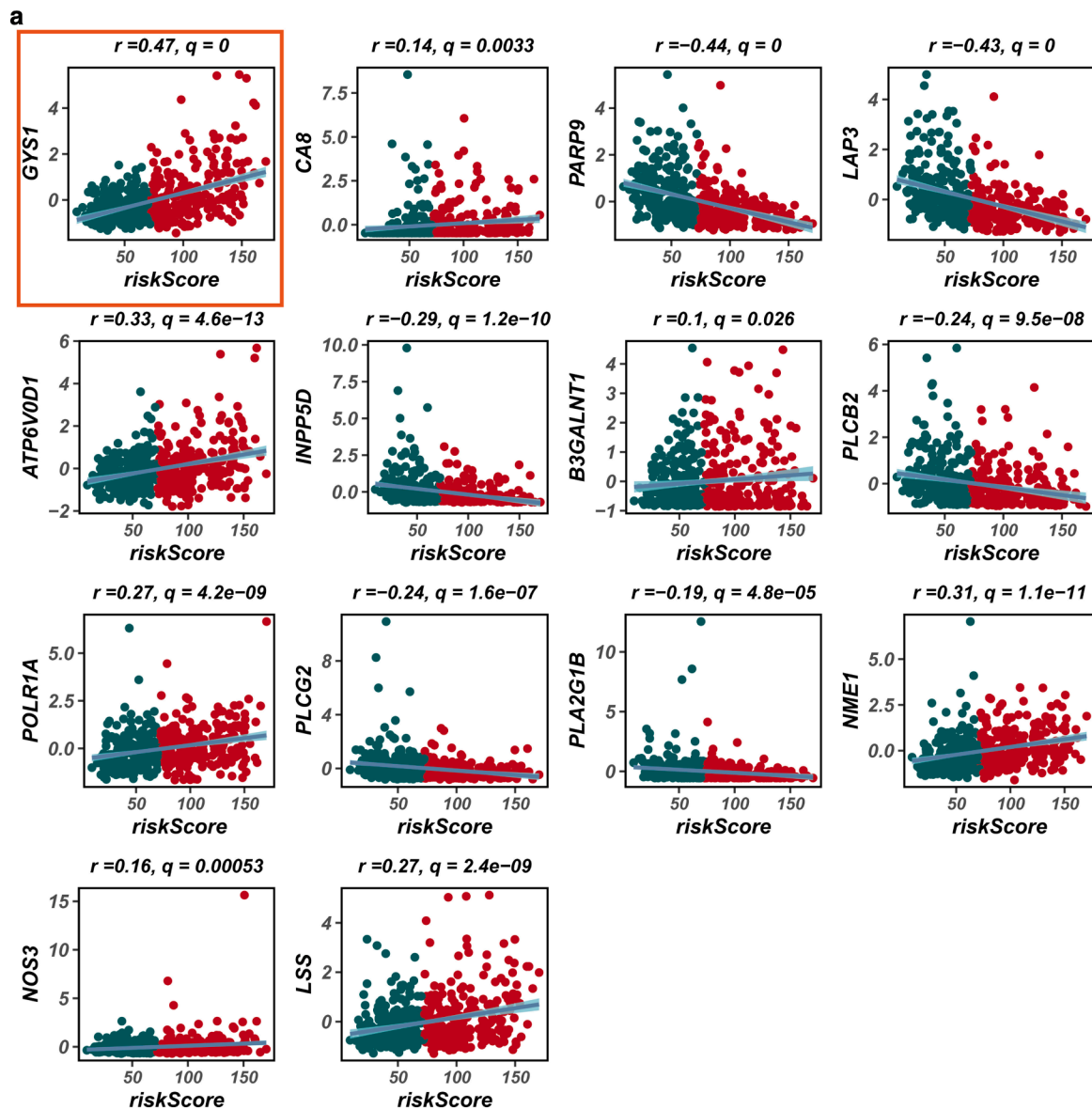
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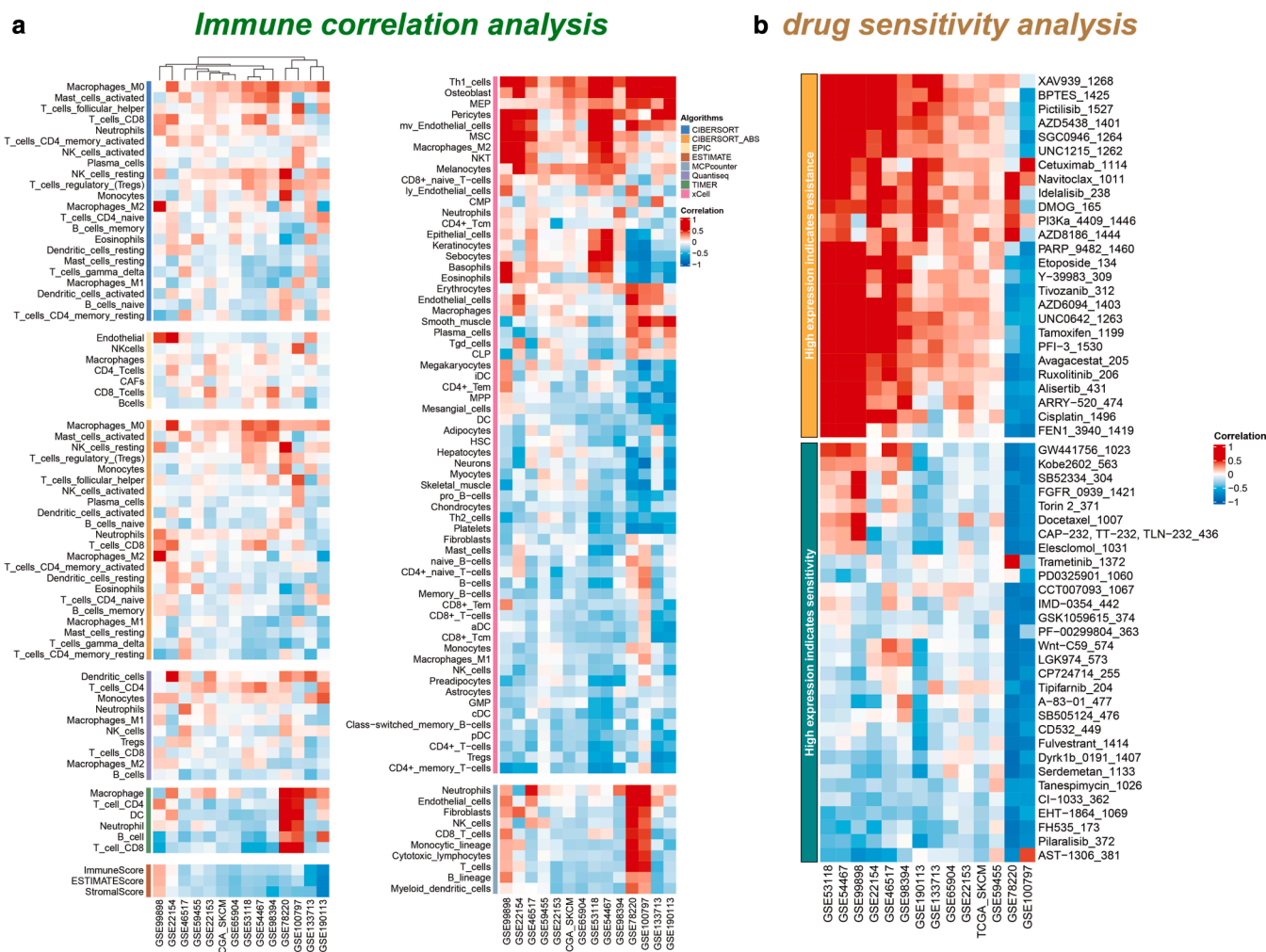
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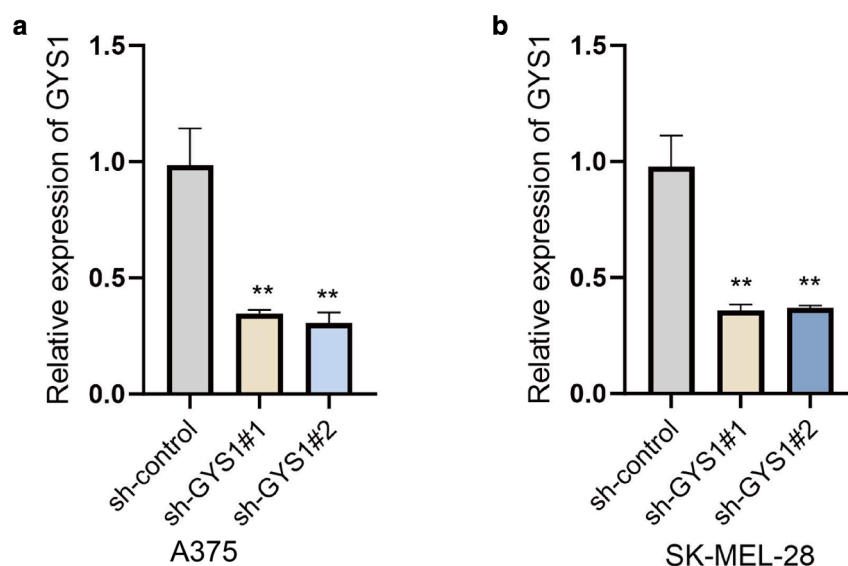
Supplementary Figure S1. Expression and prognostic significance of 114 metabolism-related pathways in melanoma. (a) Forest plot of Cox regression analysis showing 44 metabolism-related genes associated with poor prognosis ($HR > 1$) and 26 genes associated with favorable prognosis ($HR < 1$) across the combined melanoma cohort (TCGA, GSE19234, GSE22153, and GSE65904). (b) Batch effect correction visualized by PCA before and after data integration. (c) The location and differential expression of prognostic-related genes on the chromosomes; the red dots indicate genes that are upregulated in the disease group, whereas the black dots represent genes that are downregulated. (d, e) KEGG and GO enrichment analysis of prognosis-related metabolism genes, highlighting key metabolic pathways. (f) CNV landscape of metabolism-related genes across chromosomes, showing prevalent amplifications and deletions. (g) ssGSEA-based heatmap displaying high activation scores of metabolism-related genes across melanoma samples. CNV, copy number variation; GO, Gene Ontology;



Supplementary Figure S2. *GYS1* is the key gene in MRPM. (a) The coefficients and correlations of the 14 model genes in terms of their proportions in the model. (b) The glycogen concentration in melanoma tumor tissues and normal tissues was detected. It was significantly increased in the tumor tissues ($n = 3$, $**P < .01$). MRPM, metabolism-related prognostic model.

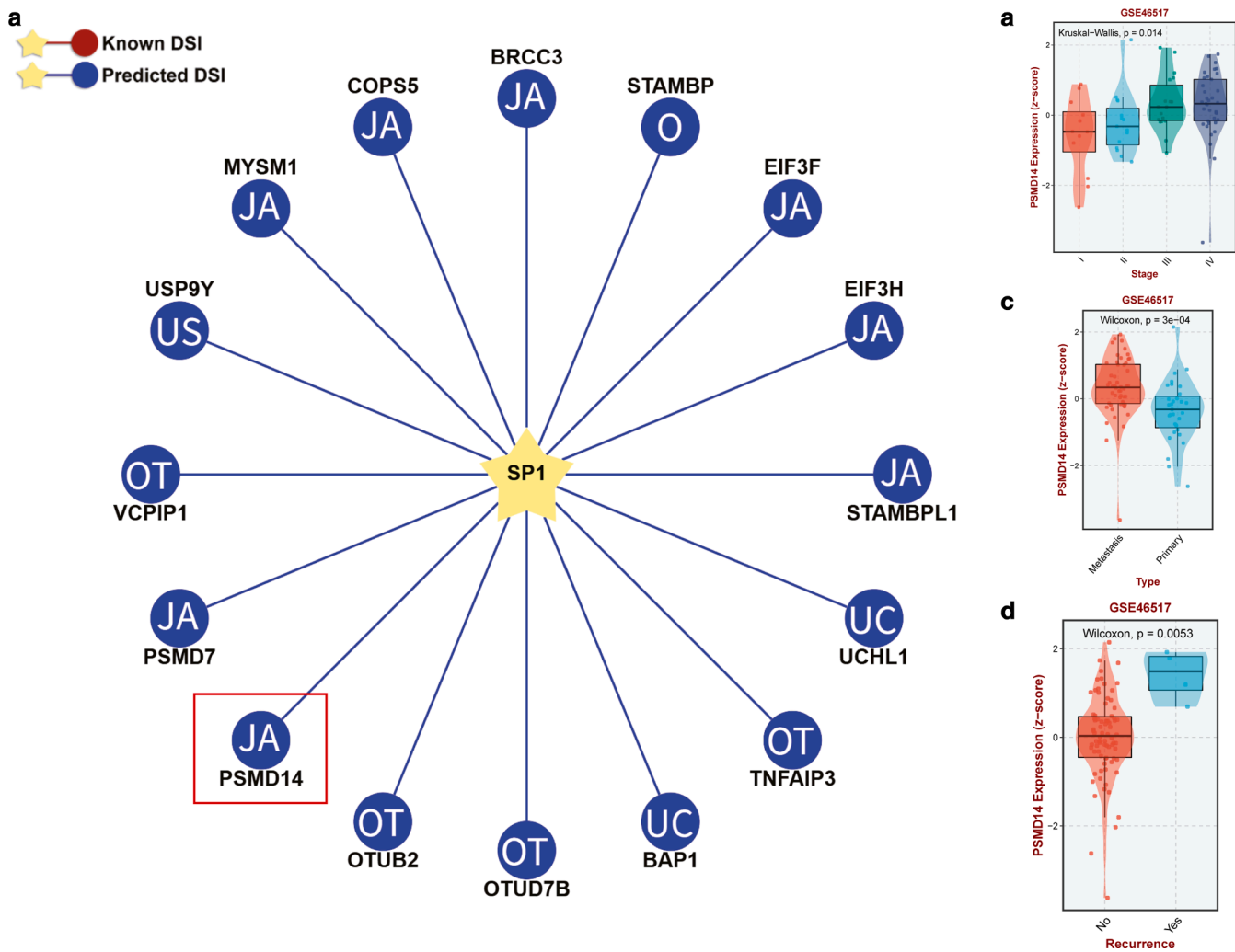


Supplementary Figure S3. The correlation between GYS1 and immune cell infiltration as well as drug sensitivity. (a) Heatmap of immune infiltration correlation. **(b)** Heatmap of drug sensitivity.



Supplementary Figure S4. PCR detection of knockdown efficiency of GYS1 in A375 and SK-MEL-28 cells.

(a) A375. (b) SK-MEL-28.



Supplementary Figure S5. PSMD14 is a potential upstream DUB of SP1. (a) The prediction results of Ubibrowser for upstream DUBs of SP1. (b–d) PSMD14 is related to the stage, metastasis, and recurrence of melanoma. DUB, deubiquitinating enzyme.

Supplementary Table S1. Importance Value of Model Genes

Gene	Importance
<i>GYS1</i>	0.0322966113849517
<i>CA8</i>	0.0219188871653145
<i>PAPR9</i>	0.0299792468437693
<i>LAP3</i>	0.0309388414579169
<i>ATP6V0D1</i>	0.0241268375700099
<i>INPP5D</i>	0.00214390509069203
<i>B3GALNT1</i>	0.0103433387531603
<i>PLCB2</i>	0.00601209130478016
<i>POLR1A</i>	0.00146231908765021
<i>PLCG2</i>	0.000479199859594475
<i>PLA2G1B</i>	0.00786790392426216
<i>NME1</i>	0.00497815389301107
<i>NOS3</i>	0.00419319397128656
<i>LSS</i>	0.00227847924581939

Supplementary Table S3. PCR Primers

Gene	Forward Primer	Reverse Primer
<i>GYS1</i>	TGGGATACCTGCAACATCGG	TTGTAGAAGTCCACGGCACC
<i>SP1</i>	GCCCTCTGACCAAGATCACT	TGGGTGACTCAATTCTGCTGC
<i>PSMD14</i>	TCACCCTGGCTTTGGTTGTT	CACAGCTCTCTCCGACAAGG
<i>ACTB</i>	TCACCATGGATGATGATATCGC	ATAGGAATCCTTCTGACCCATGC